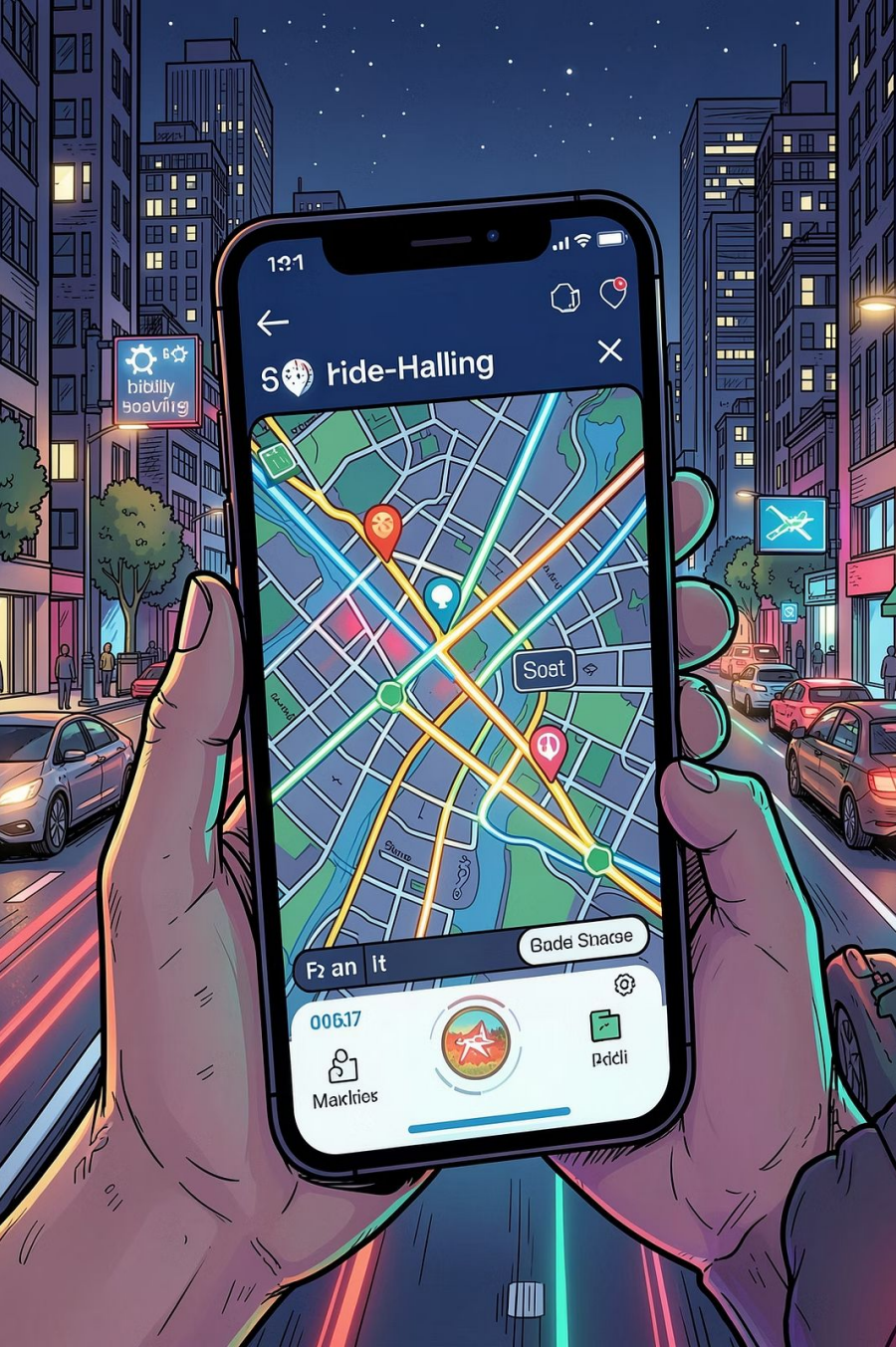




OLA Ride Booking Analysis

End-to-End Data Analytics Project using **Python**, **MySQL Workbench**, **SQL** & **Power BI** – uncovering booking trends, cancellation patterns, customer satisfaction, and operational efficiency.

Project Overview



Objective

Analyze OLA ride booking data to generate **actionable business insights** that support data-driven decision-making across key operational areas.

Tools Used

Python

MySQL

SQL

Power BI

Dataset Summary

103K

Total Rows

103,024 ride bookings in the dataset

20

Columns

Attributes spanning bookings, rides,
ratings & payments

35M

Total Booking Value

Aggregate revenue across all bookings

Key features include: **Booking_ID**, **Booking_Status**, **Customer_ID**, **Vehicle_Type**, **Booking_Value**, **Ride_Distance**, **Customer_Rating**, **Driver_Ratings**, **Payment_Method**, **Pickup_Location**, **Drop_Location**, and **Cancellation_Details**.

Missing Value Analysis

Most missing values are **business-driven** — not data quality issues — and were intentionally retained.

Column	Missing Values	Reason
V_TAT, C_TAT	39,057	Not applicable
Canceled_Rides_by_Customer	92,525	Only cancelled rides
Canceled_Rides_by_Driver	84,590	Only cancelled rides
Incomplete_Rides	39,057	Incomplete rides
Incomplete_Rides_Reason	99,098	Incomplete rides only
Payment_Method	39,057	Cancelled/Incomplete rides
Driver_Ratings	39,057	Completed rides only
Customer_Rating	39,057	Completed rides only

Exploratory Data Analysis

01

Data Loading

Imported dataset using Pandas

02

Initial Exploration

df.head(), df.info(), df.describe(), df.shape

03

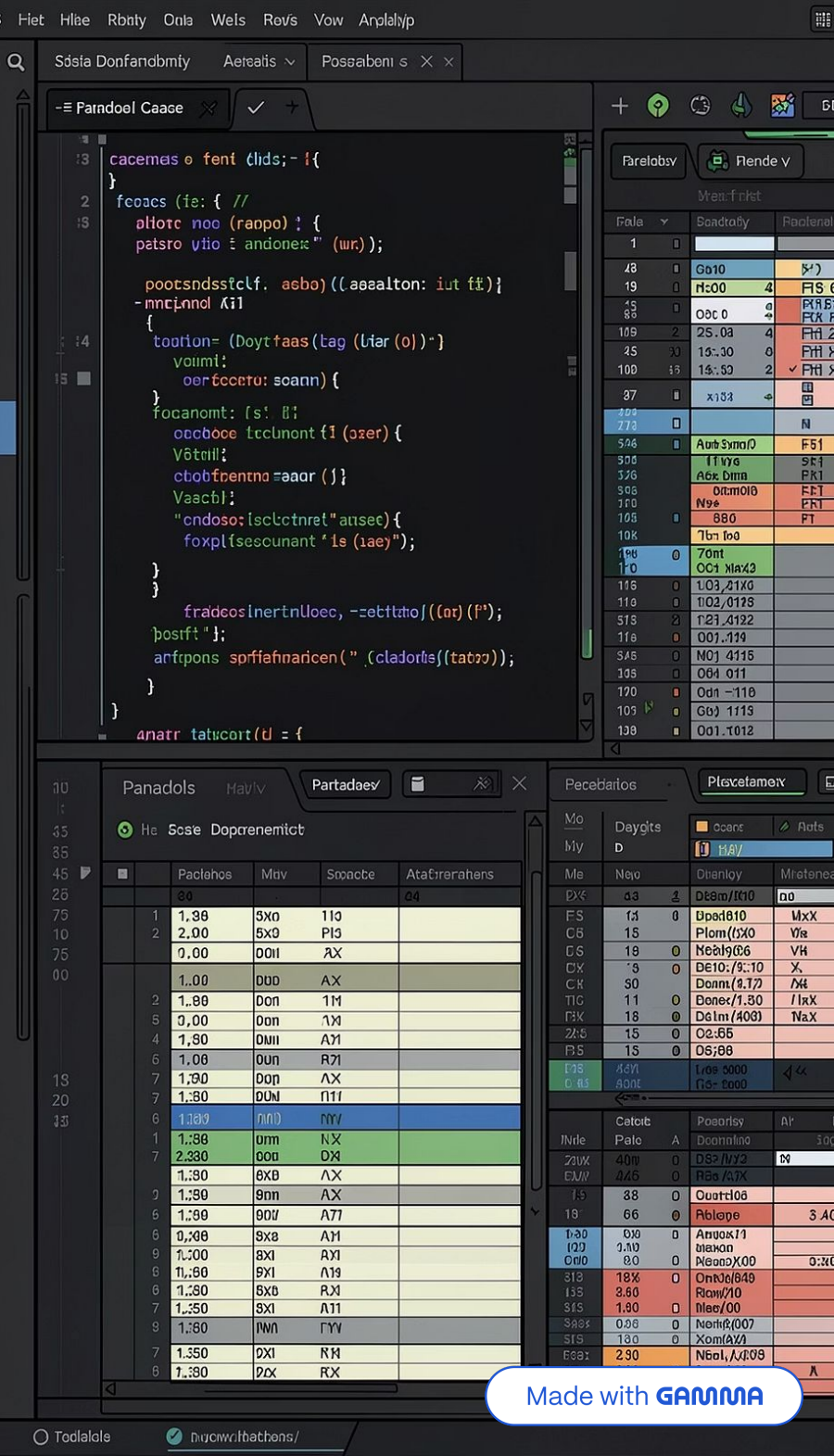
Data Cleaning

Missing value analysis, duplicate check, data type validation, column standardization

04

Validation & Integration

Data consistency check, feature validation, database integration with MySQL Workbench



Overall Dashboard

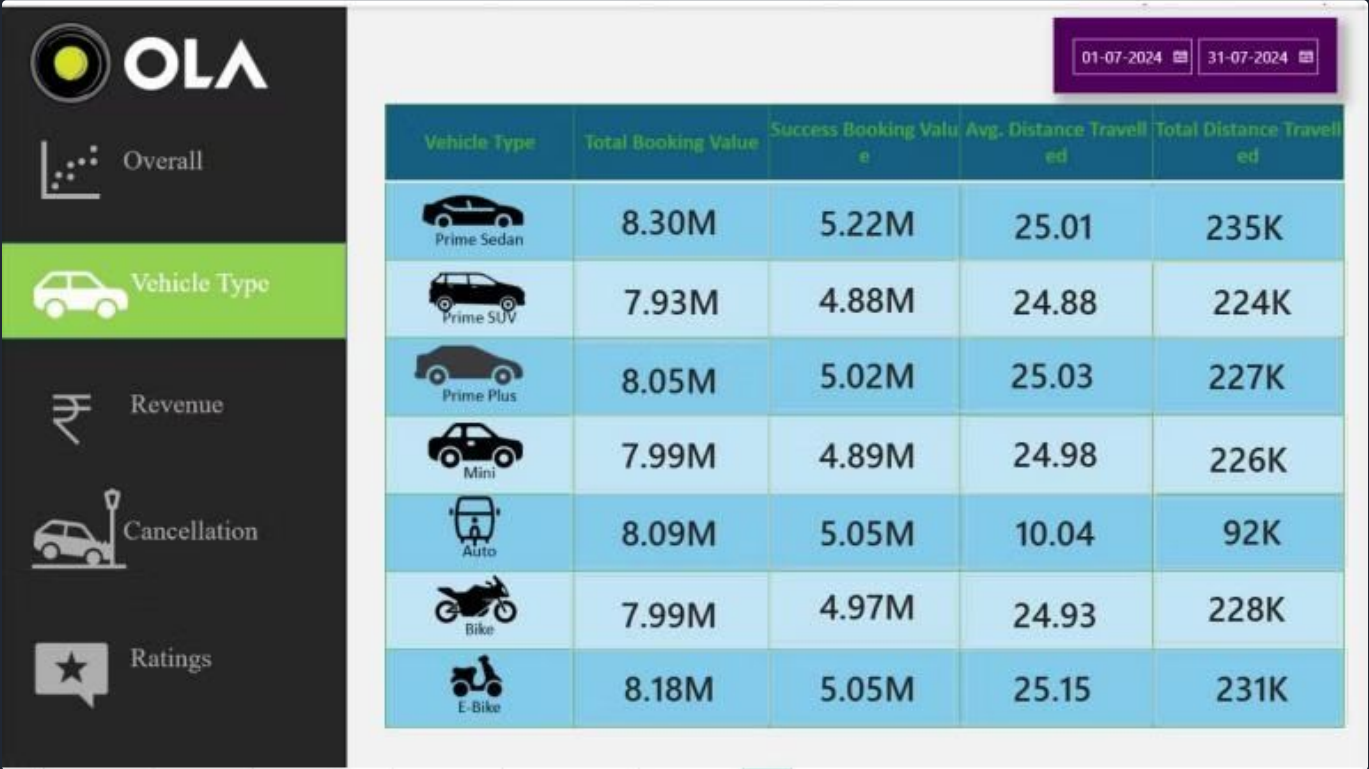


Key Metrics

- **Total Bookings:** 103,024
- **Total Booking Value:** 35M
- **Success Rate:** 63.97%
- **Canceled by Driver:** 10.19%
- **Canceled by Customer:** 9.83%
- **Driver Not Found:** 17.89%

The Overall Dashboard provides a comprehensive view of OLA's performance with booking status breakdown and ride value trends over time.

Vehicle Type Performance



Vehicle Type	Total Value	Success Value	Avg Distance
Prime Sedan	8.30M	5.22M	25.01
Prime SUV	7.93M	4.88M	24.88
Prime Plus	8.05M	5.02M	25.03
Mini	7.99M	4.89M	24.98
Auto	8.09M	5.05M	10.04
Bike	7.99M	4.97M	24.93
E-Bike	8.18M	5.05M	25.15

Revenue & Cancellation Insights



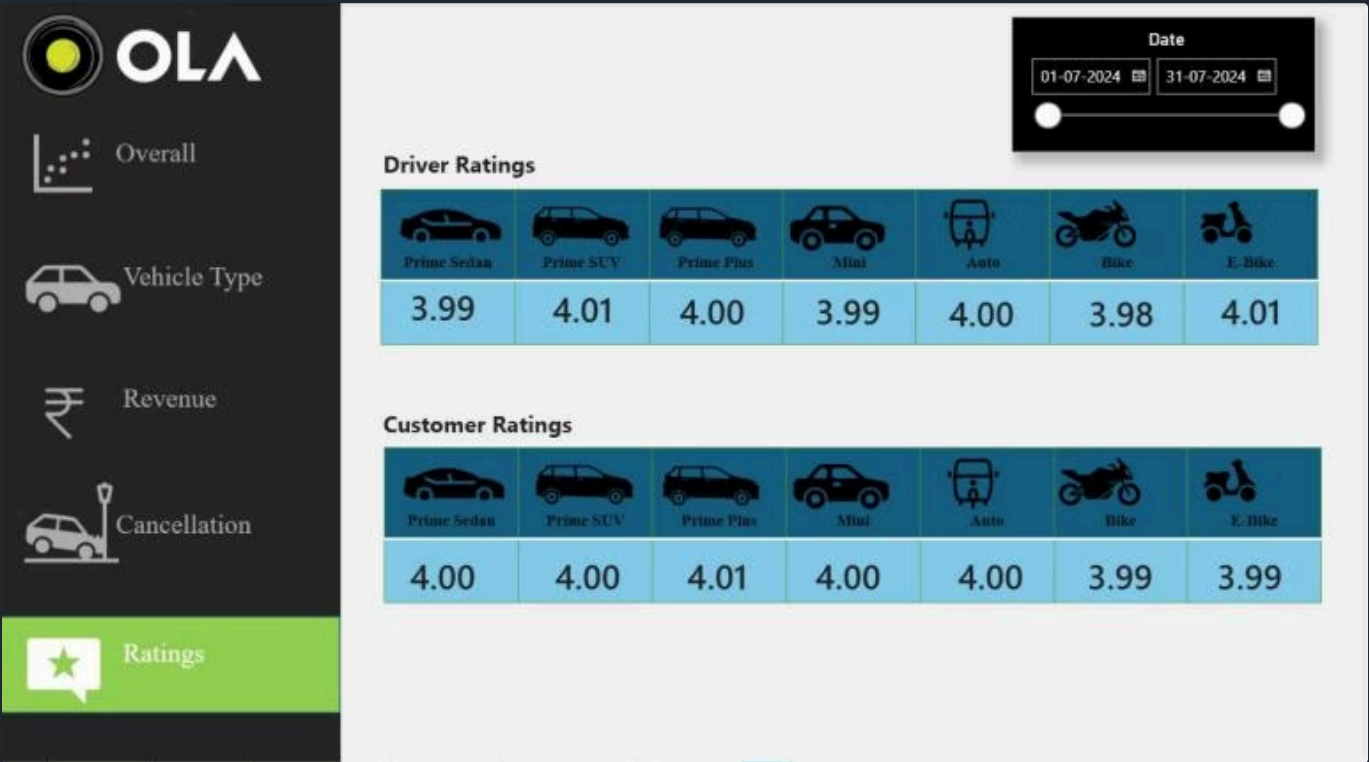
Revenue by Payment Method

- **Cash:** ~18M
- **UPI:** ~12M
- **Credit Card:** ~1M
- **Debit Card:** ~0.5M

Cancellation Highlights

- **Total Cancellations:** 28,933
- **Overall Cancellation Rate:** 17.89%
- **Top Customer Reasons:** Change of plans, Wrong address, AC not working
- **Top Driver Reasons:** Personal/car issues, Customer-related issues

Ratings Dashboard



Driver Ratings by Vehicle Type

Vehicle	Rating
Prime Sedan	3.99
Prime SUV	4.01
Prime Plus	4.00
Mini	3.99
Auto	4.00
Bike	3.98
E-Bike	4.01

Customer Ratings by Vehicle Type

Vehicle	Rating
Prime Sedan	4.00
Prime SUV	4.00
Prime Plus	4.01
Mini	4.00
Auto	4.00
Bike	3.99
E-Bike	3.99

Business Recommendations

→ Improve Driver Allocation

Reduce cancellations by optimizing driver dispatch logic.

→ Enhance ETA Prediction

Better accuracy to reduce customer-initiated cancellations.

→ Reward Top Drivers

Incentivize highly rated drivers to maintain service quality.

→ Promote High-Performing Vehicles

Focus marketing on top-performing vehicle categories.

→ Encourage Digital Payments

Cash dominates at ~18M — drive UPI and card adoption.

→ Monitor KPIs via Power BI

Track key metrics regularly through interactive dashboards.